

■ 3.5.3 Derivatives of Unknown Functions

Differentiating a known function gives an explicit result.

```
In[1]:= D[Log[x]^2, x]
Out[1]=  $\frac{2 \operatorname{Log}[x]}{x}$ 
```

Differentiating an unknown function f gives a result in terms of f' .

```
In[2]:= D[f[x]^2, x]
Out[2]= 2 f[x] f'[x]
```

Mathematica applies the chain rule for differentiation, and leaves the result in terms of f' .

```
In[3]:= D[x f[x^2], x]
Out[3]= f[x^2] + 2 x^2 f'[x^2]
```

Differentiating again gives a result in terms of f , f' and f'' .

```
In[4]:= D[%, x]
Out[4]= 6 x f'[x^2] + 4 x^3 f''[x^2]
```

When a function has more than one argument, superscripts are used to indicate how many times each argument is being differentiated.

```
In[5]:= D[g[x^2, y^2], x]
Out[5]= 2 x g^(1,0)[x^2, y^2]
```

This represents $\frac{\partial}{\partial x} \frac{\partial}{\partial x} \frac{\partial}{\partial y} g(x, y)$.

Mathematica assumes that the order in which derivatives are taken with respect to different variables is irrelevant.

```
In[6]:= D[g[x, y], x, x, y]
Out[6]= g^(2,1)[x, y]
```

You can find the value of the derivative when $x = 0$ by replacing x with 0.

```
In[7]:= % /. x->0
Out[7]= g^(2,1)[0, y]
```

$f'[x]$	first derivative of a function of one variable
$f^{(n)}[x]$	n^{th} derivative of a function of one variable
$f^{(n_1, n_2, \dots)}[x]$	derivative of a function of several variables, n_i times with respect to variable i

Output forms for derivatives of unknown functions.